

Guidelines for Effective Use

1. UVGI is line-of-sight inactivation and hence it is important to provide adequate space for objects, preventing overlap. To achieve better UVC exposure on surfaces in rooms, UVC light devices are re-positioned around objects to ensure maximum direct coverage of surfaces.
2. For proper disinfection, the UV radiation must reach a particular area with sufficient strength. UV shadows are the areas where another object blocks the UV light. A simple method to detect these areas is to place a regular A-style light bulb in the location of the UVGI device and observe which areas receive insufficient UVC radiation.
3. UV reflectivity from normal objects is very low. Some hospitals in the USA paint room walls of patients with UVC reflective paint to get better reflectivity during UVC sterilisation. A common way to compensate for poor reflectivity is to increase power and time duration of the dose.
4. UV may not sterilise inside the dirt or contamination and sterilise only their surface. Hence, before applying UVC, sanitation/disinfection of the walls should be done.
5. The amount of UVC energy delivered to kill all pathogens in room surfaces should be monitored with respect to UVC power of the UVGI luminaire and the exposure time. If the power is low, more time will be required to get a complete inactivation of all pathogens. It is recommended for all facilities to have a UVC irradiance meter to monitor the UVC dosage falling on surfaces.
6. UVGI lamps such as hot-cathode mercury vapour lamps have a life of 9000 hours, and their intensity decays over time. So, luminaire monitoring will reveal the need for replacing the lamp. Also, low-pressure mercury vapour lamps tend to have poor output at lower room temperature. Hence, it is advisable to test the output using appropriate UVC meters at normal operating temperature.
7. Facilities should regularly conduct and document the swab tests of the locations to verify the effectiveness of the UVGI protocol, as this will indicate the need for any modification of the protocol.
8. UVC radiation can damage plastics and organic material, so care should be taken to avoid using more UVC energy than required.
9. As the penetration power of UVC is limited, PPE for UVC can include a full sleeved lab coat with raised collar and tightly woven cloth to protect exposed skin and arms, along with thick nitrile gloves. A full-face shield made of UVC opaque material should be included. Persons handling UVC must be provided with all necessary training and awareness.

lenging for end-users to figure out whether the solution being invested in will be effective for their use-case or not. The worst mistake would be to invest in an ineffective solution and let down your guard.

“UVC lights for disinfection purpose being imported and sold in Indian market are basically of two types, germicidal lamps and LED types. Germicidal lamps can be used for surface disinfection purpose only and cannot be used in human presence. UVC LED lights are very popular these days but 99% of these lights are fake and do not kill any viruses. There must be some testing and cer-

tifications by authorities regarding UVC lights for disinfection purposes. Currently, there are few standards available globally for UVC devices, but the pandemic has triggered the need to regulate human safety and official standards over the growing use of UVC. While most of the standards available now are on human safety levels and protocols for UVC use, standards are yet to be defined for reliability and performance of devices being sold,” says Hardev Singh, Director, Universal Electronics.

Some of the standards related to UVC are:

- ISO has a standard for human

safety and permissible exposure from UVC (ISO 15858:2016).

- National Institute for Occupational Safety and Health certification (NIOSH), USA has specified the maximum human exposure to 254nm UVC light at 6000 microjoules/sq cm in a period of eight hours.

- Electrical safety standards for UVGI luminaries are defined by BIS 10322.

- The International Electrotechnical Commission has set up IEC62471 for photobiological safety of lamps and lamp systems, which provides guidance for the evaluation of photobiological safety of all electrically powered non-laser sources of optical radiation emitting in a spectral range of 200 to 3000nm.

- International UL 1995 standard is applied to UV lamps that are factory- and field-installed for HVAC&R equipment. ASHRAE Technical Committee (TC) 2.9 for ultraviolet air and surface treatment helps supervise safety standards that protect installers and service personnel from accidental UVC exposure from HVAC&R equipment. By applying safety features within the AHU, UL 1995, 5th edition standard seeks to remove variations in the design of the device that might lead to accidental exposure.

Comparison of Different UVC Light Sources

Table 1

Light Source for UVC	Mercury Vapour	UV LED	Xenon	Krypton Chlorine Excimer Lamp (Far-UVC)
Cost	Low (X)	High (3X)	High (2X)	Very High (4X)
Wavelength	254	275	200-300	207-222
Mercury Related Hazard	Yes	No	No	No
Construction	Fragile (Glass/Quartz)	Robust (Solid State)	Fragile (Quartz)	Semi Fragile (Quartz)
Life (Hours)	8,000 to 10,000	10,000 to 30,000	463	3,000
UVC Intensity	High Power (33%)	Very Low Power (2 to 5%)	Low Power (10%)	7-18 %
Human Exposure	Hazardous	Hazardous	Hazardous	Safe

Note: This table is for user's understanding and does not denote any fixed values